2.10 L'Hopital's Rule

Worked Solutions

1 a)

$$\lim_{x\to 2}\left(\frac{x^2-4}{x-2}\right)=\lim_{x\to 2}\left(\frac{2x}{1}\right)=4.$$

1 b)

$$\lim_{x\to 1}\left(\frac{\ln x}{x-1}\right)=\lim_{x\to 1}\left(\frac{1/x}{1}\right)=1.$$

1 c)

$$\lim_{x\to 0}\left(\frac{e^x-1}{x}\right)=\lim_{x\to 0}\left(\frac{e^x}{1}\right)=1.$$

1 d)

$$\lim_{x\to 0}\left(\frac{e^{2x}-1}{x}\right)=\lim_{x\to 0}\left(\frac{2e^{2x}}{1}\right)=2.$$

1 e)

$$\lim_{x\to 0}\left(\frac{e^x-1}{\ln(1+x)}\right)=\lim_{x\to 0}\left(\frac{e^x}{1/(x+1)}\right)=1.$$

1 f)

$$\lim_{x\to 0}\left(\frac{\ln(1+3x)}{x}\right)=\lim_{x\to 0}\left(\frac{3/(1+3x)}{1}\right)=3.$$

2 a)

$$\lim_{x\to 0}\left(\frac{\sin x}{x}\right)=\lim_{x\to 0}\left(\frac{\cos x}{1}\right)=1.$$

2 b)

$$\lim_{x\to 0}\left(\frac{1-\cos x}{x}\right)=\lim_{x\to 0}\left(\frac{\sin x}{1}\right)=0.$$

2 c)

$$\lim_{x\to 0}\left(\frac{x}{\sin x}\right)=\lim_{x\to 0}\left(\frac{1}{\cos x}\right)=1.$$

2 d)

$$\lim_{x\to 0}\left(\frac{x^2}{\sin x}\right)=\lim_{x\to 0}\left(\frac{2x}{\cos x}\right)=0.$$

2 e)

$$\lim_{x\to 0}\left(\frac{-x+\tan x}{x}\right)=\lim_{x\to 0}\left(\frac{-1+\sec^2 x}{1}\right)=\lim_{x\to 0}\left(-1+\frac{1}{\cos^2 x}\right)=1.$$

2 f)

$$\lim_{x\to 0}\left(\frac{\sin 3x}{x}\right)=\lim_{x\to 0}\left(\frac{3\cos 3x}{1}\right)=3.$$

3 a)

$$\lim_{x\to 0}\left(\frac{1-\cos x}{x^2}\right)=\lim_{x\to 0}\left(\frac{\sin x}{2x}\right)=\lim_{x\to 0}\left(\frac{\cos x}{2}\right)=\frac{1}{2}.$$

3 b)

$$\lim_{x\to 0}\left(\frac{x-\sin x}{x^3}\right)=\lim_{x\to 0}\left(\frac{1-\cos x}{3x^2}\right)=\lim_{x\to 0}\left(\frac{\sin x}{6x}\right)=\lim_{x\to 0}\left(\frac{\cos x}{6}\right)=\frac{1}{6}.$$

3 c)

$$\lim_{x\to \infty}\left(\frac{\ln x}{x}\right)=\lim_{x\to \infty}\left(\frac{1/x}{1}\right)=\lim_{x\to \infty}\left(\frac{1}{x}\right)=0.$$

3 d)

$$\lim_{x\to \infty}\left(\frac{x^2}{e^x}\right)=\lim_{x\to \infty}\left(\frac{2x}{e^x}\right)=\lim_{x\to \infty}\left(\frac{2}{e^x}\right)=0.$$

3 e)

$$\lim_{x\to 0}\left(\frac{e^{3x}-1-x}{x^2}\right)=\lim_{x\to 0}\left(\frac{3e^{3x}-1}{2x}\right)=\lim_{x\to 0}\left(\frac{9e^{3x}}{2}\right)=\frac{1}{2}.$$

3 f)

$$\lim_{x\to 0}\left(\frac{\ln(1+x)-x}{x^2}\right)=\lim_{x\to 0}\left(\frac{(x+1)^{-1}-1}{2x}\right)=\lim_{x\to 0}\left(\frac{-(x+1)^{-2}}{2}\right)=-\frac{1}{2}.$$

3 g)

$$\lim_{x\to\infty}\left(\frac{\ln x}{\ln(x+1)}\right)=\lim_{x\to\infty}\left(\frac{1/x}{1/(x+1)}\right)=\lim_{x\to\infty}\left(\frac{x+1}{x}\right)=\lim_{x\to\infty}\left(1+\frac{1}{x}\right)=1.$$

4 a)

$$\lim_{x\to 0}\left(\frac{\sinh x}{x}\right)=\lim_{x\to 0}\left(\frac{\cosh x}{1}\right)=1.$$

4 b)

$$\lim_{x\to 0}\left(\frac{\tanh x}{2x}\right)=\lim_{x\to 0}\left(\frac{\operatorname{sech}^2 x}{2}\right)=\frac{1}{2}.$$

4 c)

$$\lim_{x\to 0}\left(\frac{\sinh x-x}{4x^3}\right)=\lim_{x\to 0}\left(\frac{\cosh x-1}{12x^2}\right)=\lim_{x\to 0}\left(\frac{\sinh x}{24x}\right)=\lim_{x\to 0}\left(\frac{\cosh x}{24}\right)=\frac{1}{24}.$$

5 a)

$$\lim_{x\to 0}\left(\frac{\arcsin x}{x}\right)=\lim_{x\to 0}\left(\frac{1/\sqrt{1-x^2}}{1}\right)=1.$$

5 b)

$$\lim_{x\to 0}\left(\frac{\arctan x}{x}\right)=\lim_{x\to 0}\left(\frac{1/(1+x^2)}{1}\right)=1.$$

6 a)

They applied L'Hôpital's rule, but it wasn't in indeterminate form, because as $x\to 0$, $\cos x+1\to 2$ and $e^x+2\to 3$.

6 b)

$$\lim_{x\to 0}\left(\frac{\cos x+1}{e^x+2}\right)=\frac{2}{3}.$$

7)

$$\lim_{x\to\infty}(xe^{-2x}) = \lim_{x\to\infty}\left(\frac{x}{e^{2x}}\right) = \lim_{x\to\infty}\left(\frac{1}{e^{2x}}\right) = 0.$$

8 a)

$$x\ln\left(1+\frac{1}{x}\right) = \frac{\ln(1+1/x)}{1/x}.$$

8 b)

$$\lim_{x\to\infty}\left(\frac{5\ln(1+1/x)}{1/x}\right) = \lim_{x\to\infty}\left(\frac{-5x^{-2}/(1+1/x)}{-x^{-2}}\right) = \lim_{x\to\infty}\left(\frac{5}{1+1/x}\right) = 5.$$

9 a)

$$\lim_{x\to 0}\left(\frac{xe^x}{\sin x}\right) = \lim_{x\to 0}\left(\frac{xe^x+e^x}{\cos x}\right) = \frac{1}{1} = 1.$$

9 b)

$$\lim_{x\to 0}\left(\frac{e^x\sin 7x}{\ln(4x+1)}\right) = \lim_{x\to 0}\left(\frac{7e^x\cos 7x + e^x\sin 7x}{4/(4x+1)}\right) = \frac{7}{4}.$$

9 c)

$$\begin{aligned}\lim_{x\to 0}\left(\frac{\sin x - x\cos x}{x^3}\right) &= \lim_{x\to 0}\left(\frac{\cos x + x\sin x - \cos x}{3x^2}\right) = \lim_{x\to 0}\left(\frac{x\sin x}{3x^2}\right)\\ &= \lim_{x\to 0}\left(\frac{x\cos x + \sin x}{6x}\right) = \lim_{x\to 0}\left(\frac{-x\sin x + 2\cos x}{6}\right) = \frac{1}{3}.\end{aligned}$$

9 d)

$$\begin{aligned}\lim_{x\to 0}\left(\frac{1-\cos 5x + 3x\sin 3x}{2x^2}\right) &= \lim_{x\to 0}\left(\frac{5\sin 5x + 9x\cos 3x + 3\sin 3x}{4x}\right)\\ &= \lim_{x\to 0}\left(\frac{25\cos 5x - 27x\sin 3x + 18\cos 3x}{4}\right) = \frac{43}{4}.\end{aligned}$$

10 a)

$$\lim_{x\to\pi}\left(\frac{\sin x}{x-\pi}\right) = \lim_{x\to\pi}\left(\frac{\cos x}{1}\right) = -1.$$

10 b)

$$\lim_{x\to\pi}\left(\frac{x\cos x+\pi}{(x-\pi)^2}\right)=\lim_{x\to\pi}\left(\frac{-x\sin x+\cos x}{2(x-\pi)}\right)$$
$$=\lim_{x\to\pi}\left(\frac{-x\cos x-2\sin x}{2}\right)=\frac{\pi}{2}.$$

10 c)

$$\lim_{x\to\pi}\left(\frac{\cos(x/2)}{x\sin 2x}\right)=\lim_{x\to\pi}\left(\frac{-\frac{1}{2}\sin(x/2)}{2x\cos 2x+\sin 2x}\right)=-\frac{1}{4\pi}.$$

11 a)

$$\lim_{a\to\infty}\left(\frac{a}{e^a}\right)=\lim_{x\to\infty}\left(\frac{1}{e^a}\right)=0.$$

11 b)

Let $u=x$, $\frac{dv}{dx}=-e^{-x}$, so $\frac{du}{dx}=1$, $v=e^{-x}$. So

$$\int xe^{-x}\,dx=xe^{-x}+\int e^{-x}\,dx=xe^{-x}-\frac{1}{e^x}(+c),$$
$$\int_0^\infty xe^{-x}\,dx=\lim_{b\to\infty}\int_0^b xe^{-x}\,dx=\lim_{b\to\infty}\left[xe^{-x}-\frac{1}{e^x}\right]_0^b$$
$$=(0-0)-(0-1)=1.$$

12 a)

$$\lim_{a\to\infty}\left(\frac{\ln a}{a}\right)=\lim_{a\to\infty}\left(\frac{1/a}{1}\right)=0.$$

12 b)

Let $u=\ln x$, $\frac{dv}{dx}=x^{-2}$, so $\frac{du}{dx}=1/x$, $v=-x^{-1}$. So

$$\int\frac{\ln x}{x^2}\,dx=-x^{-1}\ln x-\int(-x^{-1}\times 1/x)\,dx=-\frac{1}{x}\ln x+\int x^{-2}\,dx$$
$$=-\frac{1}{x}\ln x-\frac{1}{x}(+c),$$
$$\int_1^\infty\frac{\ln x}{x^2}\,dx=\lim_{b\to\infty}\left[-\frac{1}{x}\ln x-\frac{1}{x}\right]_1^b=(0-0)-(-0-1)=1.$$

13 a)

$$\frac{d}{dx}[\ln(\cos x)]=\frac{1}{\cos x}\times(-\sin x)=-\tan x.$$

13 b)

$$\lim_{x\to 0}\left(\frac{\ln(\cos x)}{x^2}\right)=\lim_{x\to 0}\left(\frac{-\tan x}{2x}\right)=\lim_{x\to 0}\left(\frac{-\sec^2 x}{2}\right)=-\frac{1}{2}.$$

14 a)

$$\frac{d}{dx}[\ln(\cosh x)]=\frac{1}{\cosh x}\times\sinh x=\tanh x.$$

14 b)

$$\lim_{x\to 0}\left(\frac{\ln(\cosh x)-7x^2}{x^2}\right)=\lim_{x\to 0}\left(\frac{\tanh x-14x}{2x}\right)=\lim_{x\to 0}\left(\frac{\operatorname{sech}^2 x-14}{2}\right)=\frac{1-14}{2}=-\frac{13}{2}.$$

15 a)

$$\lim_{x\to 0^+}(x\ln x)=\lim_{x\to 0^+}\left(\frac{\ln x}{1/x}\right)=\lim_{x\to 0^+}\left(\frac{1/x}{-1/x^2}\right)=\lim_{x\to 0^+}(-x)=0.$$

15 b)

$\ln x$ is only defined for $x>0$ (in the real numbers). For $x<0$, $\ln x$ is not real, so the expression $x\ln x$ is not defined. Therefore, we must take the limit as $x\to 0^+$ (from the right) only.

16)

$$s(t)=\frac{e^t-1}{t},\qquad s'(t)=\frac{te^t-(e^t-1)}{t^2}=\frac{te^t-e^t+1}{t^2}.$$

$$\begin{aligned}\lim_{t\to 0}\left(\frac{te^t-e^t+1}{t^2}\right)&=\lim_{t\to 0}\left(\frac{te^t+e^t-e^t}{2t}\right)=\lim_{t\to 0}\left(\frac{te^t}{2t}\right)\\&=\lim_{t\to 0}\left(\frac{te^t+e^t}{2}\right)=\frac{1}{2}.\end{aligned}$$

Therefore velocity $=\frac{1}{2}\,\mathrm{ms}^{-1}$.

17 a)

$$\ln(1+t)=t-\frac{1}{2}t^2+\frac{1}{3}t^3-\frac{1}{4}t^4+\cdots.$$

So $\ln(1+t)-t+\frac{1}{2}t^2\approx t-\frac{1}{2}t^2+\frac{1}{3}t^3-t+\frac{1}{2}t^2=\frac{1}{3}t^3$.

17 b)

$$\lim_{x\to 0}f(x)=\lim_{x\to 0}\left(x^{-3}\left(\ln(1+x)-x+\frac{1}{2}x^2\right)\right)\approx\lim_{x\to 0}\left(x^{-3}\frac{1}{3}x^3\right)=\frac{1}{3},\quad\text{as req.}$$

17 c)

$$\lim_{x\to 0}\left(\frac{\ln(1+x)-x+\frac{1}{2}x^2}{x^3}\right)=\lim_{x\to 0}\left(\frac{(1+x)^{-1}-1+x}{3x^2}\right)$$
$$=\lim_{x\to 0}\left(\frac{-(1+x)^{-2}+1}{6x}\right)=\lim_{x\to 0}\left(\frac{2(1+x)^{-3}}{6}\right)=\frac{1}{3},\quad \text{as req.}$$

17 d)

L'Hôpital's rule determines the value of the limit, whilst the Maclaurin expansion describes the behaviour near the point. In this particular case, the Maclaurin expansion identifies the dominant term to be $\frac{1}{3}x^3$, shows how the function approaches the limit, and provides a local approximation with error terms.

18)

Let $y=(1+\frac{1}{x})^x$. So

$$\ln y=\ln\left[(1+\frac{1}{x})^x\right]=x\ln\left(1+\frac{1}{x}\right).$$

So

$$\lim_{x\to\infty}\ln y=\lim_{x\to\infty}\left(x\ln\left(1+\frac{1}{x}\right)\right).$$

Rewrite in indeterminate form:

$$\lim_{x\to\infty}\left(\frac{\ln(1+1/x)}{1/x}\right).$$

Apply L'Hôpital's rule:

$$\lim_{x\to\infty}\left(\frac{\ln(1+1/x)}{1/x}\right)=\lim_{x\to\infty}\left(\frac{-1/x^2}{(1+1/x)(-1/x^2)}\right)=\lim_{x\to\infty}\left(\frac{1}{1+1/x}\right)=1.$$

So $\lim_{x\to\infty}(\ln y)=1$, which means that $\ln y\to 1$, so $y\to e$.

$$\therefore\lim_{x\to\infty}(1+\frac{1}{x})^x=e.$$